Chapter 5: Convergence Concepts

Part I — Statistics Foundation for ML

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Abstract

A single estimate is one number; what makes statistics possible is knowing what happens to that number as the data pile up. This chapter is about the limits of random quantities. Two results carry most of the weight: the *law of large numbers*, which says the sample mean converges to the true mean, and the *central limit theorem*, which says the error of that sample mean is normal almost regardless of where the data came from. Together they are why an average is trustworthy and why the normal distribution turns up everywhere. We state them, watch them happen in code, prove them from two elementary inequalities, and close with the handful of tools — Slutsky's theorem, the continuous mapping theorem, the delta method — that the rest of the course quietly relies on.

Contents

1 The law of large numbers and the central limit theorem 1

2 Why they are true 2

3 Modes of convergence and the tools built on them 4

1 The law of large numbers and the central limit theorem

Let $X_1, X_2, \dots$ be independent and identically distributed, with mean $\mu = E[X_i]$ and finite variance $\sigma^2 = \mathrm{Var}(X_i)$. The object we care about is the **sample mean** $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Its own mean and variance follow from linearity:

$$E[\bar{X}_n] = \mu, \qquad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

The mean stays put while the variance shrinks like $1/n$, so the distribution of $\bar{X}_n$ should pile up tighter and tighter around μ. Making "piles up around" precise is what a mode of convergence is for.

Definition 1.1 (Convergence in probability)**.** A sequence Y_n *converges in probability* to Y, written $Y_n \xrightarrow{p} Y$, if for every $\varepsilon > 0$,

$$P\big(|Y_n - Y| > \varepsilon\big) \longrightarrow 0 \quad \text{as } n \to \infty.$$

Theorem 1.1 (Weak law of large numbers)**.** If $X_1, \dots, X_n$ are iid with mean μ and finite variance, then $\bar{X}_n \xrightarrow{p} \mu$.

The law of large numbers is the guarantee that estimation works at all: collect enough data and the sample mean is as close to the truth as you like. But it says nothing about *how* close,

or about the shape of the leftover error. That is the central limit theorem, and it needs a second notion of convergence — one about distributions rather than values.

Definition 1.2 (Convergence in distribution)**.** $Y_n \xrightarrow{d} Y$ if $F_{Y_n}(y) \to F_Y(y)$ at every point y where the limiting cdf F_Y is continuous.

Theorem 1.2 (Central limit theorem)**.** If $X_1, \dots, X_n$ are iid with mean μ and finite variance σ^2, then

$$\sqrt{n}\,(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2), \qquad \text{equivalently} \qquad \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

The remarkable part is what the statement does *not* assume: nothing about the shape of the X_i beyond a finite variance. Uniform, exponential, a lopsided coin — whatever the data distribution, the standardized sample mean forgets it and turns into the same bell curve. The sample size launders the shape away.

That universality is easy to doubt and easy to check. The code below takes three deliberately un-normal sources — a flat one, a skewed one, and a two-point one — and reports the mean and variance of their standardized sample means:

```
import random, math

def standardized_means(sampler, mu, sigma, n, reps, seed=0):
    rng = random.Random(seed)
    z = []
    for _ in range(reps):
        xbar = sum(sampler(rng) for _ in range(n)) / n
        z.append((xbar - mu) / (sigma / math.sqrt(n)))      # -> N(0, 1)
    return z

sources = {                                        # three very different shapes
    "Uniform(0,1)":    (lambda r: r.random(), 0.5, 1/math.sqrt(12)),
    "Exponential(1)": (lambda r: -math.log(1 - r.random()), 1.0, 1.0),
    "Bernoulli(0.2)": (lambda r: 1.0 if r.random() < 0.2 else 0.0, 0.2, 0.4),
}
for name, (samp, mu, sd) in sources.items():
    z = standardized_means(samp, mu, sd, n=40, reps=20000)
    mean = sum(z) / len(z)
    var  = sum((zi - mean)**2 for zi in z) / len(z)
    print(f"{name:14s} -> mean {mean:+.3f}, var {var:.3f}")    # ~ 0 and ~ 1
        for all
```

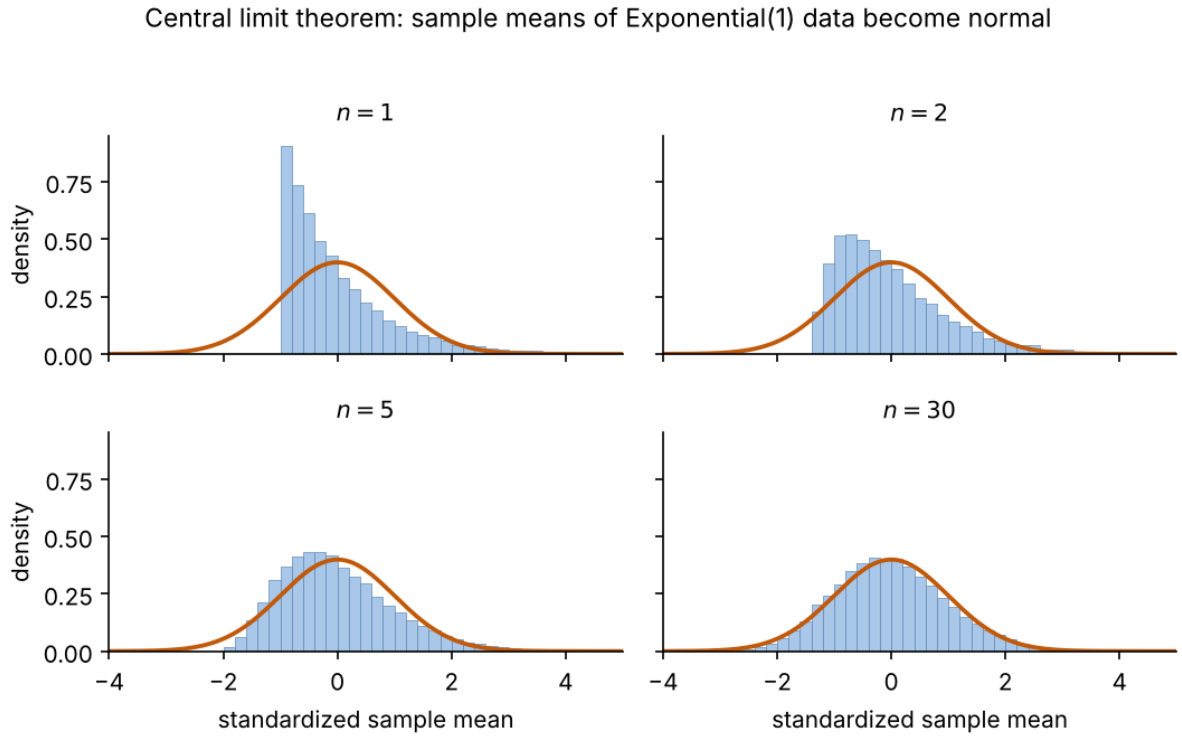
All three print a mean near 0 and a variance near 1 — the first two moments of $N(0,1)$ — no matter how un-bell-shaped the source was. The shape itself is easiest to see by watching a single skewed source as n grows:

At $n = 1$ the standardized "mean" is just the raw Exponential(1), sharply right-skewed. By $n = 5$ the skew is fading, and by $n = 30$ the histogram and the $N(0,1)$ curve are on top of each other. Nothing about the exponential survives the averaging.

2 Why they are true

Both theorems come from controlling how much probability can sit far from the mean. Two inequalities do all the work, and the first needs almost nothing.

Theorem 2.1 (Markov's inequality)**.** If $Y \geq 0$ and $a > 0$, then $P(Y \geq a) \leq E[Y]/a$.



Proof. Since $Y \geq 0$, we have $a\,\mathbf{1}\{Y \geq a\} \leq Y$ (the left side is either 0 or a, and when it is a then $Y \geq a$). Taking expectations, $a\,P(Y \geq a) \leq E[Y]$, which rearranges to the claim. □

Theorem 2.2 (Chebyshev's inequality)**.** For any random variable with mean μ and variance σ^2, and any $\varepsilon > 0$,

$$P\big(|X - \mu| \geq \varepsilon\big) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Apply Markov's inequality to the non-negative variable $Y = (X - \mu)^2$ with $a = \varepsilon^2$:

$$P\big((X - \mu)^2 \geq \varepsilon^2\big) \leq \frac{E[(X - \mu)^2]}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2}.$$

The event $(X - \mu)^2 \geq \varepsilon^2$ is the same as $|X - \mu| \geq \varepsilon$. □

The weak law now falls out in one line.

Proof of the weak law of large numbers. The sample mean has variance σ^2/n, so Chebyshev's inequality gives, for any $\varepsilon > 0$,

$$P\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \leq \frac{\sigma^2/n}{\varepsilon^2} = \frac{\sigma^2}{n\,\varepsilon^2} \longrightarrow 0.$$ □

The central limit theorem is deeper, but a clean proof runs through the moment generating function.

Theorem 2.3 (CLT via the MGF, sketch)**.** Standardize: let $Y_i = (X_i - \mu)/\sigma$, so $E[Y_i] = 0$, $\mathrm{Var}(Y_i) = 1$, with MGF $m(t) = 1 + \frac{t^2}{2} + o(t^2)$. Write $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Y_i$. By independence,

$$M_{Z_n}(t) = \Big[m\big(t/\sqrt{n}\big)\Big]^n = \Big[1 + \frac{t^2}{2n} + o\big(1/n\big)\Big]^n \longrightarrow e^{t^2/2},$$

which is the MGF of $N(0, 1)$. Convergence of MGFs implies convergence in distribution, so

$Z_n \xrightarrow{d} N(0,1)$.

The limit $e^{t^2/2}$ depends only on the first two moments of the Y_i — mean 0 and variance 1 — which is exactly why every finite-variance distribution lands on the same normal curve.

3 Modes of convergence and the tools built on them

We have used two modes of convergence; there are four worth naming, from strongest to weakest. $Y_n \to Y$ **almost surely** ($\xrightarrow{a.s.}$) if $P(\lim_n Y_n = Y) = 1$; **in mean square** if $E|Y_n - Y|^2 \to 0$; **in probability**; and **in distribution**. The implications go one way,

$$\xrightarrow{a.s.} \implies \xrightarrow{p} \implies \xrightarrow{d}, \qquad \text{mean square} \implies \xrightarrow{p},$$

and none of the arrows reverses — except that convergence in distribution *to a constant* does imply convergence in probability. (The "weak" in the weak law is this distinction: the *strong* law upgrades the same conclusion to almost-sure convergence.)

Three tools let these limits be combined, and they are used constantly once estimators get plugged into one another.

Theorem 3.1 (Slutsky)**.** If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$ for a constant c, then $X_n + Y_n \xrightarrow{d} X + c$, $X_n Y_n \xrightarrow{d} cX$, and $X_n / Y_n \xrightarrow{d} X/c$ (for $c \neq 0$).

This is what lets you replace an unknown σ in $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ by a consistent estimate $\hat{\sigma}$ and keep the same normal limit — the basis of every t-statistic and confidence interval.

Theorem 3.2 (Continuous mapping)**.** If g is continuous and $X_n \xrightarrow{d} X$, then $g(X_n) \xrightarrow{d} g(X)$ (the same holds for $\xrightarrow{p}$ and $\xrightarrow{a.s.}$).

Theorem 3.3 (Delta method)**.** If $\sqrt{n}\,(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$, then

$$\sqrt{n}\,\big(g(\hat{\theta}) - g(\theta)\big) \xrightarrow{d} N\big(0,\ g'(\theta)^2\,\sigma^2\big).$$

Sketch. A first-order Taylor expansion gives $g(\hat{\theta}) = g(\theta) + g'(\theta)(\hat{\theta} - \theta) + R_n$, where the remainder R_n is negligible because $\hat{\theta} \xrightarrow{p} \theta$. Multiplying by $\sqrt{n}$, the leading term is $g'(\theta) \cdot \sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} g'(\theta)\, N(0, \sigma^2) = N(0, g'(\theta)^2 \sigma^2)$, and Slutsky absorbs the remainder. □

The delta method is how the asymptotic normality of an estimator (for instance the maximum-likelihood estimator of Chapter 6) passes to any smooth function of it. When the terms being summed are independent but *not* identically distributed — weighted sums, regression residuals — the central limit theorem still holds under the **Lindeberg condition**, which asks only that no single term contributes a non-vanishing share of the total variance.

Summary

The sample mean converges to the truth (law of large numbers, $\bar{X}_n \xrightarrow{p} \mu$) and its error is normal whatever the data distribution (central limit theorem, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$). Both follow from Markov's and Chebyshev's inequalities and the moment generating function. Four modes of convergence (almost sure, mean square, probability, distribution) are linked by one-way implications, and Slutsky's theorem, the continuous mapping theorem and the delta method are the tools that carry these limits through the estimators built in

the rest of the course.